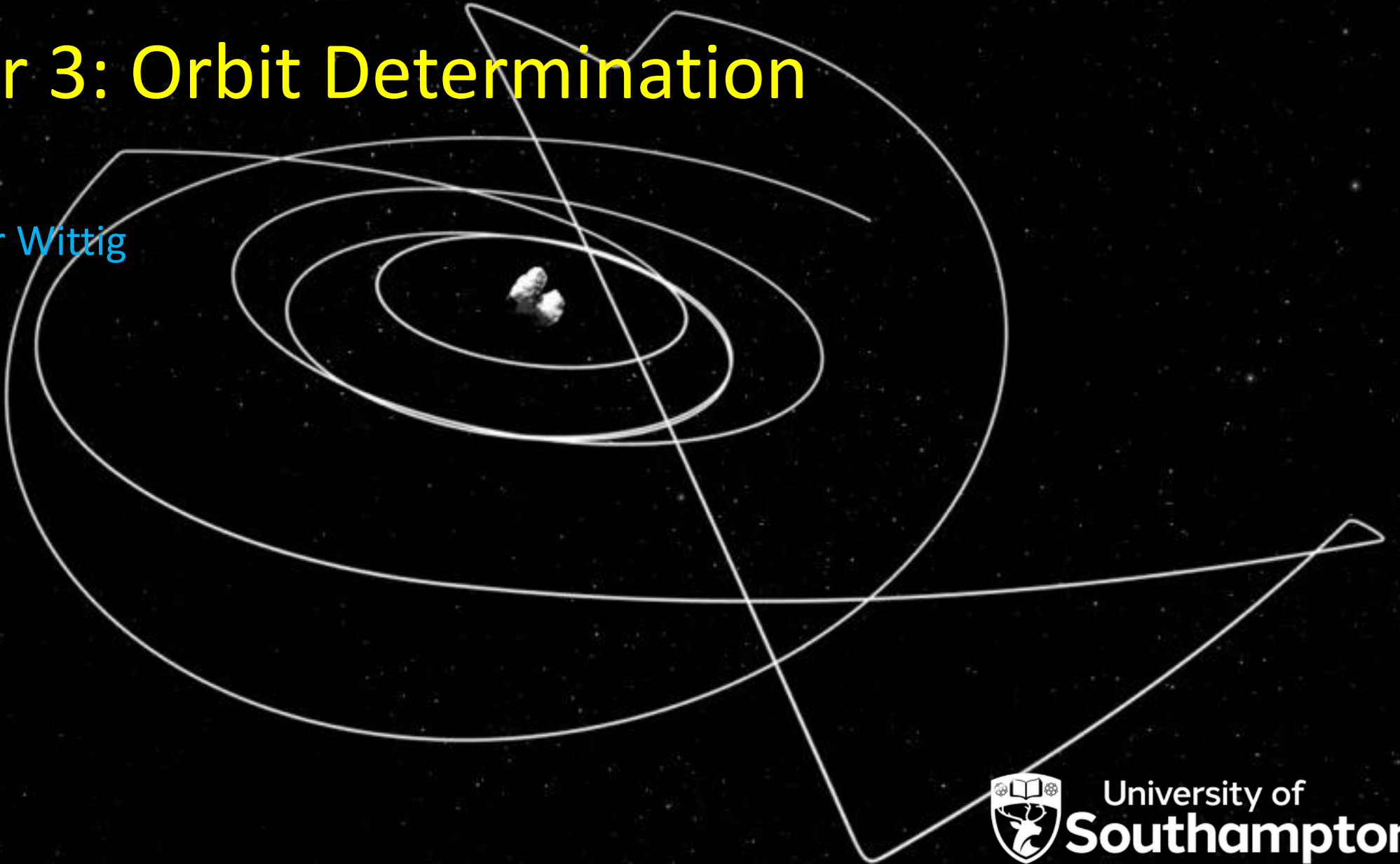


# Orbital Mechanics (SESA6076)

## Chapter 3: Orbit Determination

Dr. Alexander Wittig



- Any questions on previous weeks content?

- **Introduction**
- **Chapter 1: In 20.000 km, turn right. Orbital Manoeuvres**
  - Coplanar Manoeuvres
  - Non-Coplanar Transfers
- **Chapter 2: That's mildly disturbing Orbital Perturbations & Propagation**
  - Non-Uniform Gravity Field
  - Third-Body Perturbation
  - Atmospheric Drag
  - Solar Radiation Pressure
  - Orbit Propagation Methods
- **Chapter 3: Is it a plane? Is it a bird? Orbit Observation & Determination**
  - Observation Techniques
  - Preliminary Orbit Determination
- **Chapter 4: Waltzing Matilda Relative Orbital Motion**
  - Relative Orbital Motion
  - Linearization & State Transition Matrix
- **Chapter 5: To boldly go... Interplanetary Trajectories & Optimization**
  - Lambert's Problem
  - Patched Conics
  - Gravity-assisted fly-by trajectories
- **Chapter 6: Torn between two bodies Restricted Three-Body Problem**
  - N-Body Problem
  - Circular Restricted Three-Body Problem
  - Natural Dynamics around Libration Points
- **Chapter 7: You spin me right round Attitude Dynamics**
  - Attitude representation
  - Attitude Dynamics
- **Chapter 8: In the spin cycle Attitude Motion**
  - Torque-Free Motion
  - Attitude Maneuvers
- **Summary and Revision**

- Identify advantages and disadvantages of key observation techniques for orbit determination
- Understand the process of preliminary orbit determination
- Perform orbit determination given different observations
  - 3 positions
  - 2 positions + time
  - 3 angles
- Perform Gibb's method
- Understand Lambert's problem and its solution
- Outline Gauss' method
- Describe Lagrange F, G functions







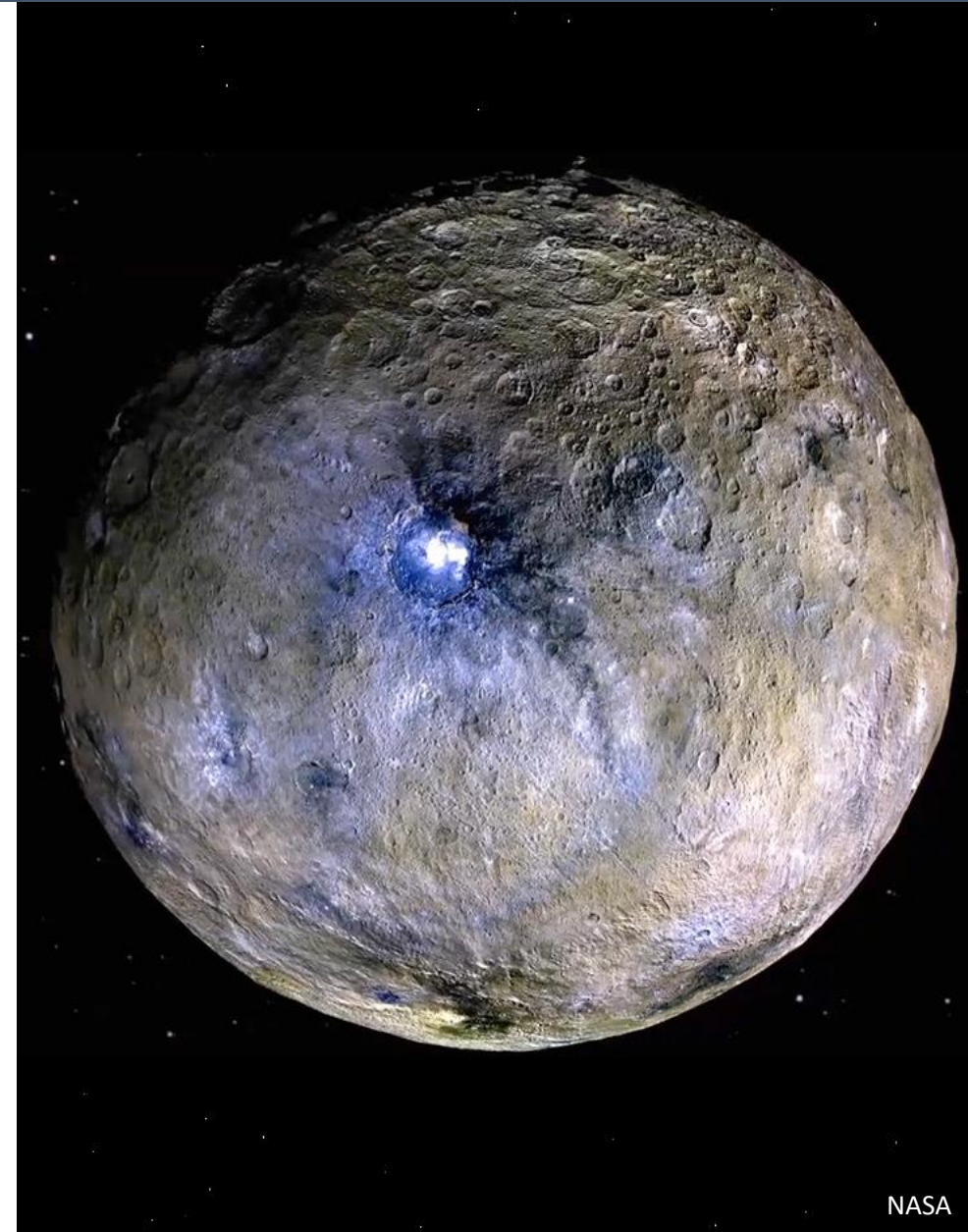




- 1000 new **Near Earth Objects (NEOs)** discovered every year
- Tracking existing objects
- Newly launched satellites / rocket bodies
- Space debris catalogue

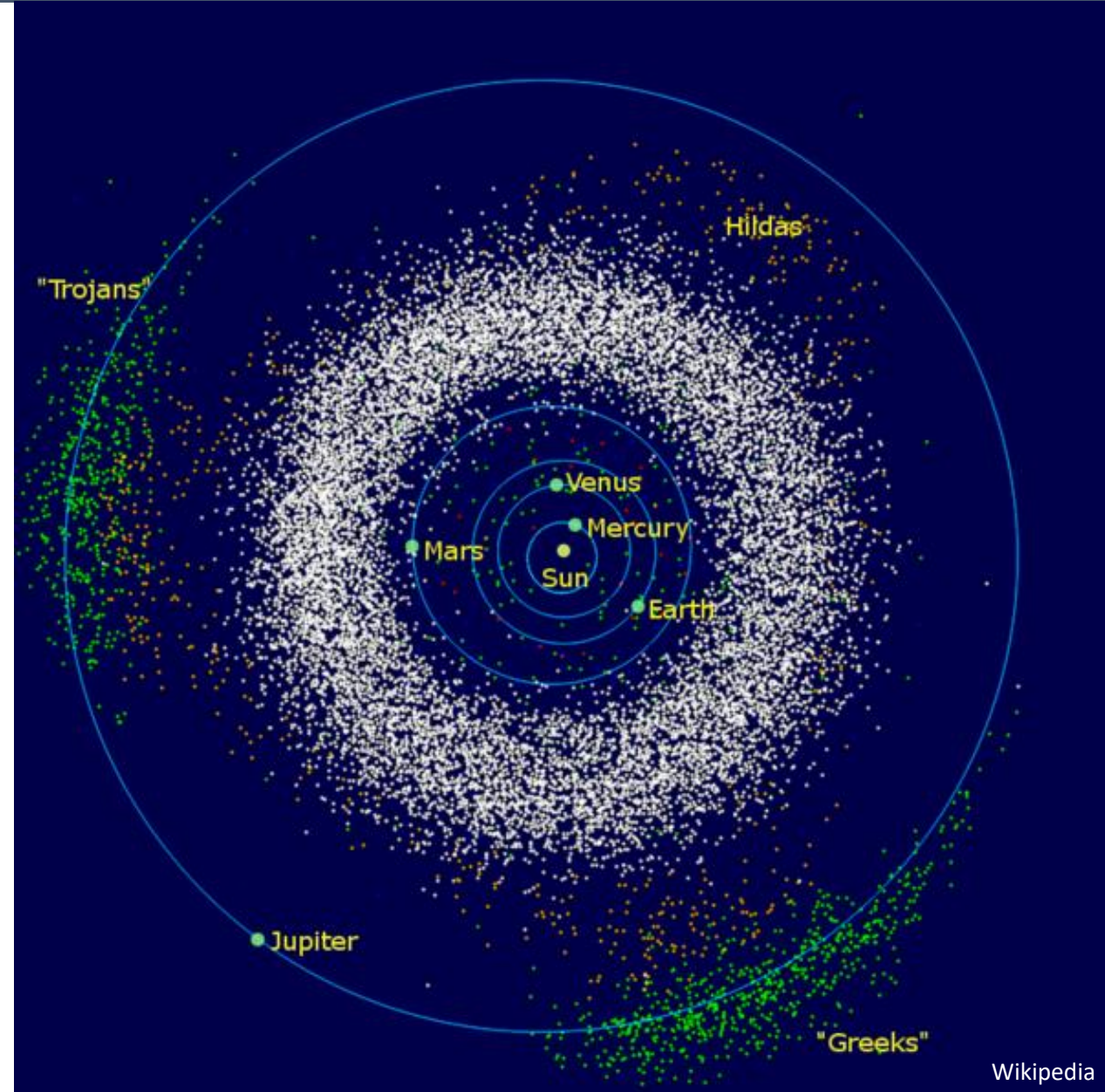


- In 1801 Giuseppe Piazzi discovers first asteroid, **Ceres**, while making star map
- Distinction between (dwarf) planet and asteroid is fluent (c.f. **Pluto**, **Vesta**)
- Asteroid belt between Mars and Jupiter contains thousands of asteroids
- Comets similar to asteroids but thought to be less solid (hence the tail)





- Small objects
  - 22% mass of Pluto
  - 4% mass of Moon
- Trojans and Greeks
  - Dynamically separated
  - Effect of gravity of Jupiter and Sun
- Not only asteroid accumulation
  - NEOs
  - Kuiper belt
  - Oort cloud





Corbis Images

## So how are celestial bodies found?

### 1. Observation

- Observe sky, noting angles & times
- Check image against catalogue



## So how are celestial bodies found?

### 1. Observation

- Observe sky, noting angles & times
- Check image against catalogue

### 2. (Preliminary) Orbit Determination

- Use observation to derive orbital elements
- Predict future observations
- Check



Celestron

# Observation Techniques



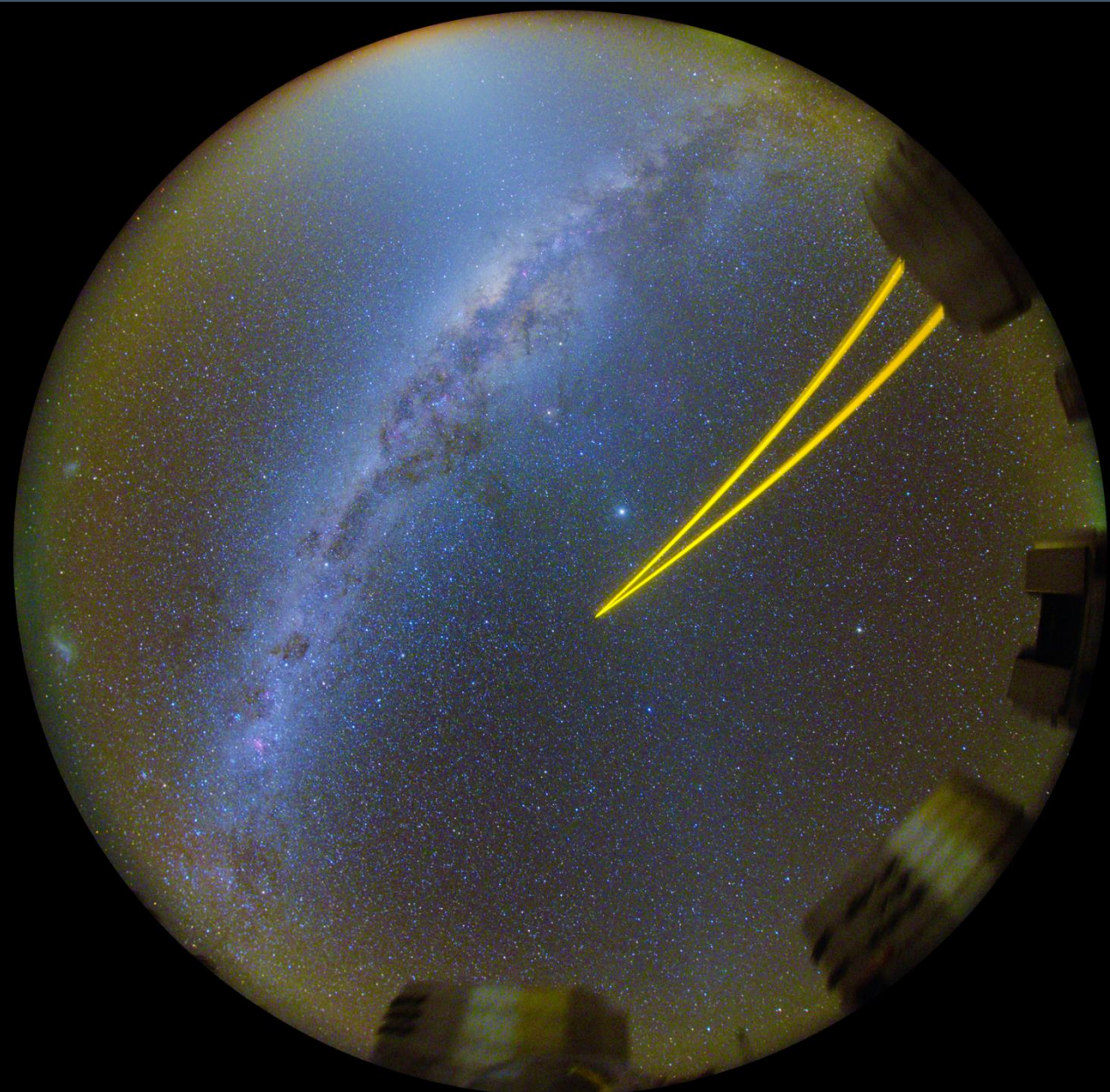


ESO



Celestron







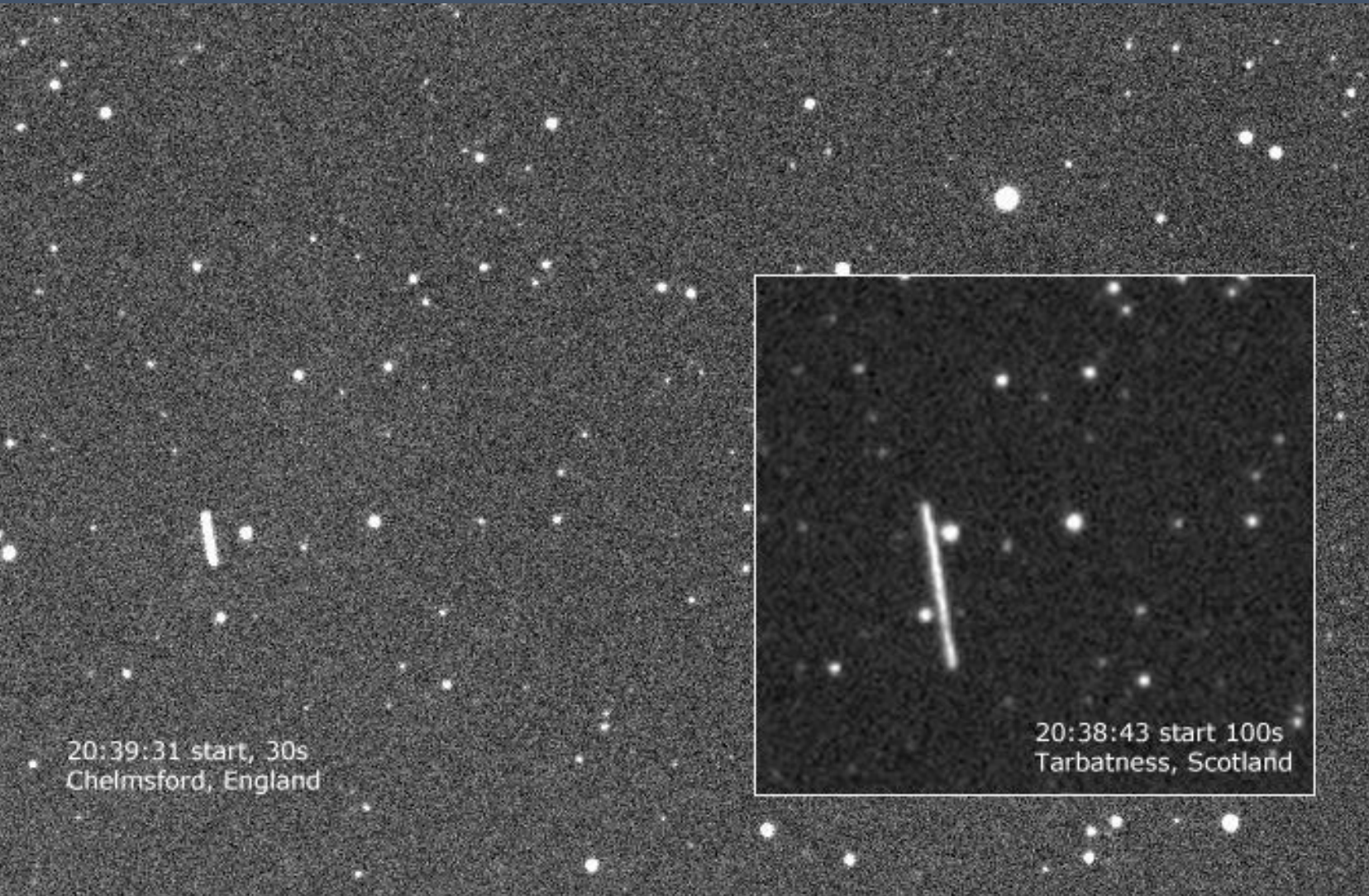
The asteroid/comet 596 Scheila photographed remotely by Australian amateur astronomer Joseph Brimacombe using a telescope in New Mexico.

The glowing fuzz or coma around the asteroid is visible in larger amateur telescopes.

Credit: Joseph Brimacombe



Allows optical observation of asteroid/comet features



CLOSE APPROACH OF (357439) 2004 BL86  
Jan 26, 2015

From 20:10 - 00:15, 381 frames on 11  
field centres, 30s exposures, FoV  
72'x54' (SXV-H9 + Megrez 72).

Compared with individual frame from  
Tarbatness, Scotland.

Credit: Nick James, Denis Buczynski

Unknown asteroid ephemeris requires tracking of stars.



ESA XMM X-Ray telescope



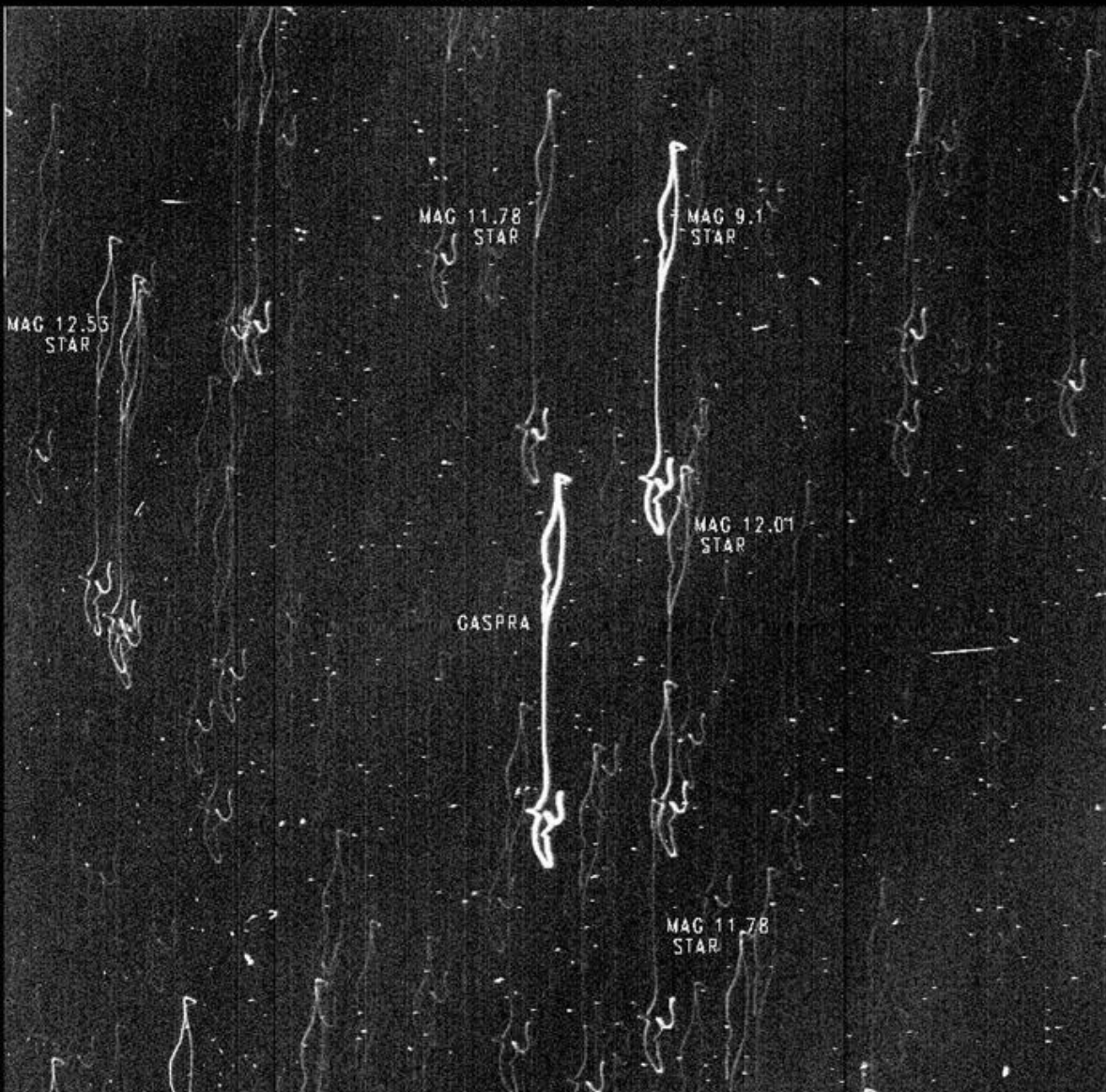
## Advantages:

- No atmospheric distortion
- No absorption of IR & UV

## Disadvantages:

- Costly
- Hard to maintain & upgrade
- Size limitations

OPNAV 4 Gaspra and Star Trails  
October 13, 1991



This NASA/JPL time-exposure picture of the asteroid Gaspra and background stars is one of four optical navigation images made by Galileo's imaging system to improve knowledge of Gaspra's location for the spacecraft flyby.

The 26-second exposure was made October 13, 1991, when Galileo was about 11 million kilometers (7 million miles) from the asteroid.

The camera was deliberately slewed about 0.14 degrees to stretch the asteroid and star images into streaks. Information from this and other navigation images was used to design trajectory correction maneuvers to place the spacecraft in the best path for its observations of Gaspra.

Sometimes slew is intentionally introduced to enhance images





Hubble/ESA

## Career:

- Doctorate in Astronomy (University of Chicago, 1949)
- Chief of Astronomy (NASA, 1959)
- Hubble program scientist (NASA, 1980)

## Achievements:

- “Mother of the Hubble”
- First woman to hold executive position at NASA

<https://www.nasa.gov/content/goddard/what-did-hubble-see-on-your-birthday>

- High resolution images
- Large spectrum
  - Visible
  - Radio
  - X-Ray
- Allow accurate angle / angular rate
- Passive
- No range / range rate

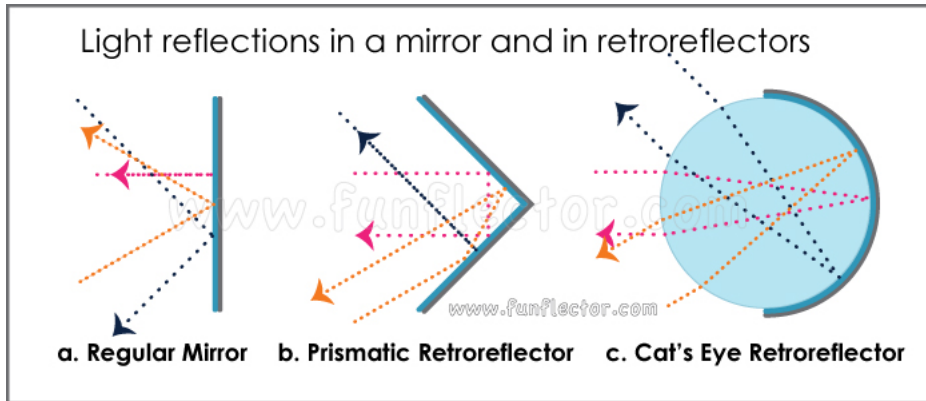
Lovell Radio Telescope, Cheshire East, UK  
(Wikipedia)



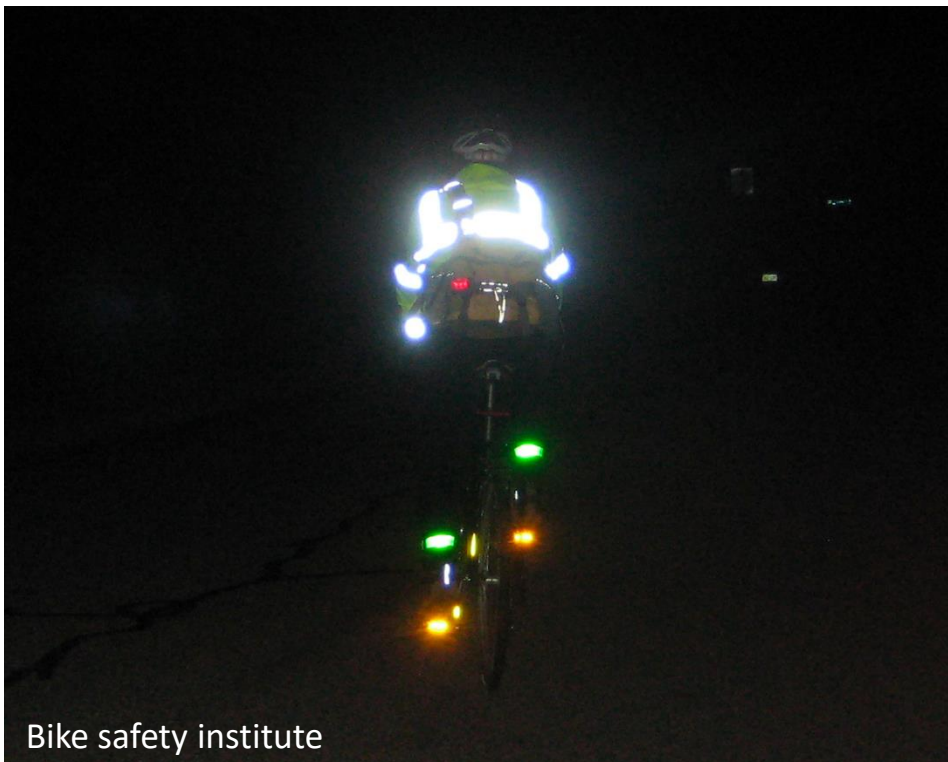


- Similar to “optical” telescope in cm wavelength range
- Active: send signal, observe reflection
- Allows scanning passive objects
- Surface information
- Range, range rate from signal runtime





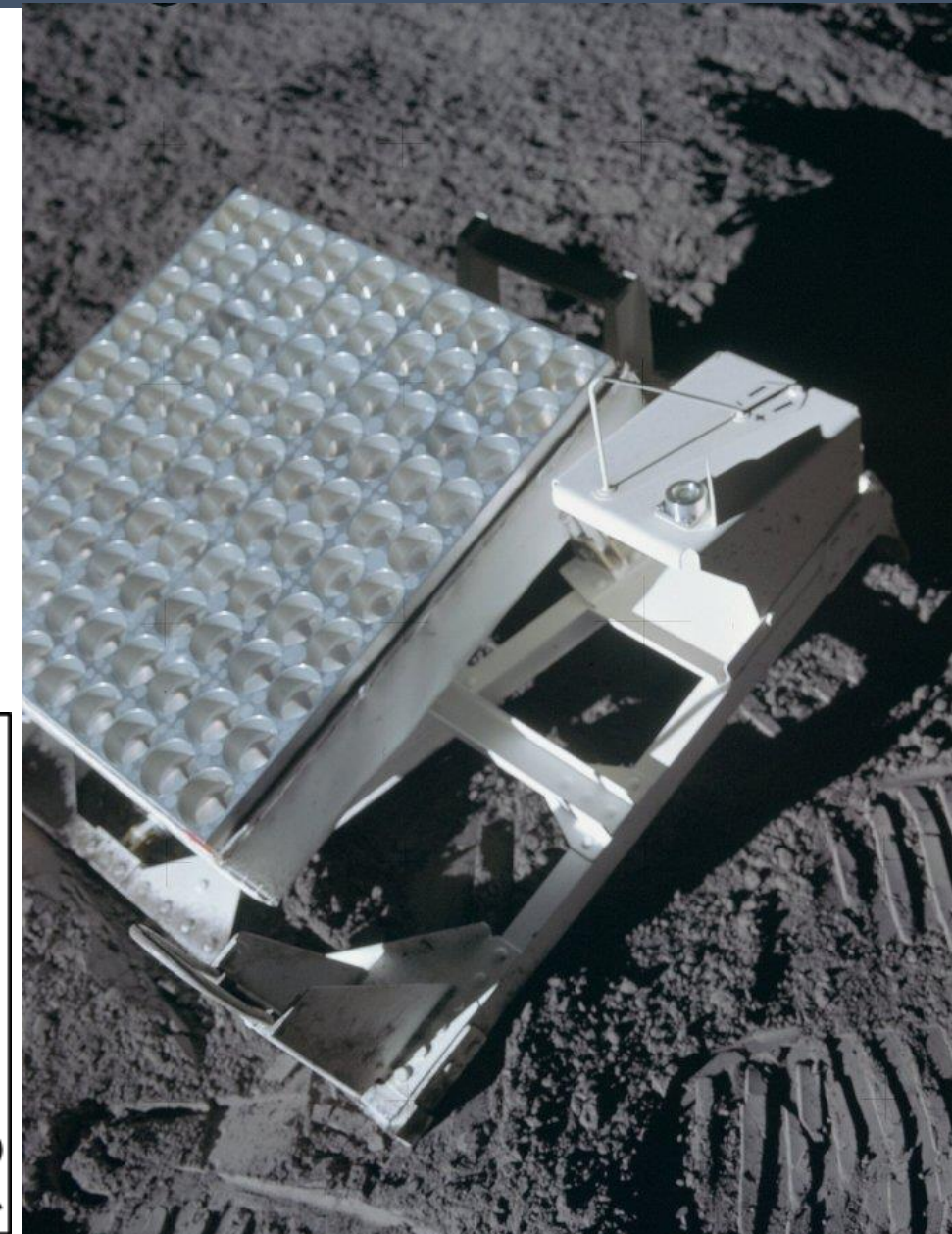
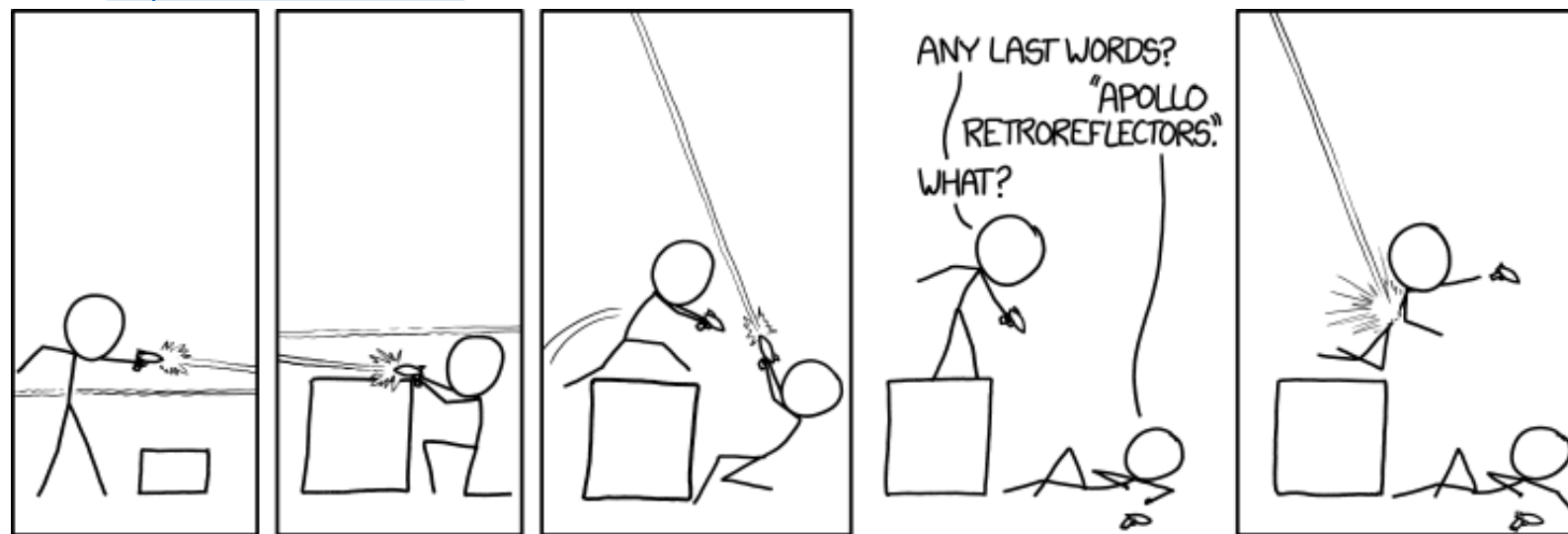
- Similar idea but using (near) visible laser
- Active: send signal, observer reflection
- Requires retro-reflectors
- Highly accurate range, range rate (mm accuracy)
- Requires very accurate pointing





- Apollo Retro-reflectors
- High accuracy Moon ephemeris
- Space based geodesy
- High accuracy Earth location
- Tectonic shifts

XKCD <https://xkcd.com/1441/>



## Raw observation data

### ■ Optical

- Tracklets
- Allow derivation of angles & angular rates
- Near Earth to really far away  
(Near Earth to entire visible universe)
- Passive

### ■ Radar

- Range & range rate estimates
- Poor estimates of angles & angular rates
- Near Earth + Solar system

### ■ Laser

- Highly accurate (mm) range & range rate estimates
- Requires precise orbit location
- Distances up to Moon



# Observation Data

# Preliminary Orbit Determination

Johannes Kepler's tables of planetary motion

Marcij Motus Planetarum Anno 1617.

| Initi. | Gregorian | h       | i    | 24      | ♈    | ♉       | ♊    | ♋        | ♌      | ♍       | ♎     | ♏       | ♐     | ♑       | ♒    | ♓       | ♈    |
|--------|-----------|---------|------|---------|------|---------|------|----------|--------|---------|-------|---------|-------|---------|------|---------|------|
|        |           | Longi.  | Lat. | Longi.  | Lat. | Longi.  | Lat. | Longi.   | Lat.   | Longi.  | Lat.  | Longi.  | Lat.  | Longi.  | Lat. | Longi.  | Lat. |
|        |           | Gr. Mi. | 2    | Gr. Mi. | 2    | Gr. Mi. | 2    | Gr. Mi.  | 2      | Gr. Mi. | 2     | Gr. Mi. | 2     | Gr. Mi. | 2    | Gr. Mi. | 2    |
| 19     | 1         | 6.17    | 15   | 23.24   | 11   | 24.52   | 15   | 11.12.   | 3      | 17.58   | 5     | 26.49   | 29    | 21.8    | 4.57 | 46      |      |
| 20     | 2         | 6.23    | 15   | 23.35   | 11   | 24.32   | 15   | 12.12.   | 8      | 19.9    | 1     | 28.22   | 40    | 4.39    | 4.22 | 42      |      |
| 21     | 3         | 6.28    | 14   | 23.46   | 11   | 24.13   | 10   | 13.12.10 | 20.21  | 4       | 0.11  | 52      | 18.34 | 1.32    | 39   |         |      |
| 22     | 4         | 6.34    | 14   | 23.57   | 11   | 23.54   | 8    | 14.12.10 | 21.32  | 7       | 1.27  | 7       | 2.57  | 2.26    | 36   |         |      |
| 23     | A. Lat.   | 6.40    | 14   | 24.8    | 11   | 23.35   | 6    | 15.12.   | 6      | 22.44   | 11    | 2.58    | 24    | 17.42   | 1.10 | 33      |      |
| 24     | 6         | 6.45    | 14   | 24.19   | 11   | 23.17   | 3    | 16.12.   | 0      | 23.55   | 14    | 4.13    | 46    | 2.40    | 0.13 | 30      |      |
| 25     | 7         | 6.51    | 13   | 24.30   | 11   | 22.59   | 1    | 17.11.53 | 25.6   | 10      | 5.22  | 2       | 17.48 | 1.35    | 27   |         |      |
| 26     | 8         | 6.57    | 13   | 24.42   | 12   | 22.42   | 59   | 18.11.42 | 26.17  | 27      | 6.22  | 15      | 2.57  | 2.51    | 23   |         |      |
| 27     | 9         | 7.3     | 13   | 24.53   | 12   | 22.25   | 57   | 19.11.30 | 27.28  | 25      | 7.13  | 25      | 17.59 | 3.54    | 20   |         |      |
| 28     | 10        | 7.9     | 13   | 25.4    | 12   | 22.9    | 54   | 20.11.16 | 28.39  | 26      | 7.59  | 34      | 2.44  | 4.40    | 17   |         |      |
| 1      | 11        | 7.15    | 13   | 25.15   | 12   | 21.53   | 52   | 21.11.1  | 29.50  | 33      | 8.42  | 41      | 17.8  | 5.7     | 14   |         |      |
| 2      | A. Ind.   | 7.22    | 12   | 25.26   | 12   | 21.38   | 50   | 22.10.43 | 1.0    | 37      | 9.1   | 47      | 1.6   | 5.15    | 11   |         |      |
| 3      | 13        | 7.28    | 12   | 25.36   | 12   | 21.24   | 48   | 23.10.24 | 2.11   | 40      | 9.57  | 51      | 14.24 | 5.4     | 7    |         |      |
| 4      | 14        | 7.34    | 12   | 25.47   | 12   | 21.12   | 45   | 24.10.1  | 3.21   | 44      | 10.18 | 53      | 27.16 | 4.41    | 4    |         |      |
| 5      | 15        | 7.41    | 12   | 25.58   | 12   | 21.0    | 43   | 25.9.36  | 4.32   | 48      | 10.24 | 54      | 9.51  | 4.2     | 1    |         |      |
| 6      | 16        | 7.47    | 12   | 26.8    | 12   | 20.49   | 41   | 26.9.10  | 5.42   | 52      | 10.20 | 55      | 22.13 | 3.13    | 58   |         |      |
| 7      | 17        | 7.54    | 12   | 26.18   | 12   | 20.39   | 39   | 27.8.41  | 6.52   | 55      | 10.12 | 55      | 4.25  | 2.16    | 55   |         |      |
| 8      | 18        | 8.1     | 11   | 26.28   | 13   | 20.29   | 36   | 28.8.0   | 8.2    | 59      | 9.52  | 56      | 16.31 | 1.14    | 52   |         |      |
| 9      | A. Palm   | 8.7     | 11   | 26.37   | 13   | 20.20   | 34   | 29.7.18  | 9.12   | 3       | 9.27  | 55      | 28.28 | 0.10    | 48   |         |      |
| 10     | 20        | 8.13    | 11   | 26.47   | 13   | 20.11   | 32   | 0.6.53   | 10.22  | 6       | 8.56  | 55      | 10.22 | 0.55    | 45   |         |      |
| 11     | 21        | 8.20    | 11   | 26.57   | 13   | 20.3    | 30   | 1.6.53   | 11.32  | 10      | 8.18  | 54      | 22.13 | 1.57    | 42   |         |      |
| 12     | 22        | 8.26    | 11   | 27.6    | 13   | 19.55   | 27   | 2.5.16   | 12.41  | 14      | 7.37  | 51      | 4.5   | 2.54    | 39   |         |      |
| 13     | 23        | 8.33    | 11   | 27.15   | 13   | 19.48   | 25   | 3.4.57   | 13.51  | 17      | 6.53  | 48      | 16.1  | 3.44    | 36   |         |      |
| 14     | 24        | 8.40    | 10   | 27.24   | 13   | 19.41   | 23   | 4.4.13   | 15.0   | 21      | 6.5   | 44      | 28.0  | 4.24    | 33   |         |      |
| 15     | A. Ma     | 8.47    | 10   | 27.33   | 13   | 19.35   | 21   | 5.3.26   | 16.9   | 25      | 5.15  | 37      | 10.6  | 4.53    | 29   |         |      |
| 16     | A. Paf.   | 8.54    | 10   | 27.42   | 14   | 19.32   | 18   | 6.2.35   | 17.18  | 28      | 4.24  | 29      | 22.2  | 5.9     | 26   |         |      |
| 17     | 27        | 9.1     | 10   | 27.50   | 14   | 19.31   | 16   | 7.1.45   | 18.27  | 32      | 3.32  | 21      | 4.54  | 5.10    | 23   |         |      |
| 18     | 28        | 9.8     | 10   | 27.59   | 14   | 19.30   | 14   | 8.0.49   | 19.36  | 36      | 2.41  | 12      | 17.35 | 4.56    | 20   |         |      |
| 19     | 29        | 9.15    | 10   | 28.8    | 14   | 19.30   | 12   | 8.59.53  | 20.45  | 39      | 1.51  | 1       | 0.35  | 4.26    | 17   |         |      |
| 20     | 30        | 9.22    | 9    | 28.17   | 14   | 19.30   | 9    | 9.58.56  | 21.53  | 43      | 1.150 | 13.54   | 1.42  | 1.1     |      |         |      |
| 1      | 21        | 9.30    | 9    | 28.26   | 14   | 19.31   | 7    | 10.57.58 | 23.147 | 47      | 0.1   | 36      | 27.3  | 2.45    |      |         |      |

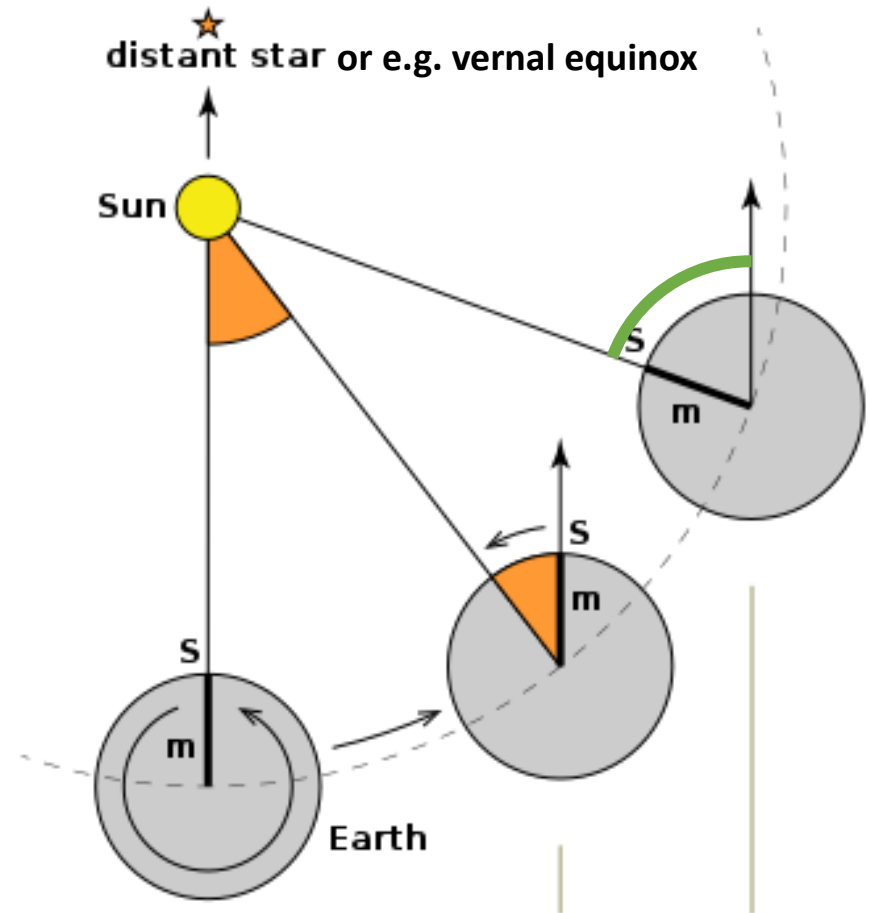
Die 3. mane ☾ nondum erat inter cor & cervicem Leonis, vixit per jam transiverat.  
Die 8. Marcij inter Zwerla & Helmeseth, occidit Luna habens a sinistra Mercurium, paulo altioremo imo  
luo margine; & videbatur linea ex Mercurio extremis Lunæ cornibus parallela incidens, incidere in pat-  
tem Lunæ obcuram, sic ut aliquotam diametri partem interciperet, forte quartam. Distabat a prop. ori-  
mine minus diametro Lunæ. Die 18. H. 7. linea per cornua ☽, videbatur quodammodo ultra Martem.

1. Combine observations to estimate orbit
  - Different times
  - Different locations
  - Different methods
2. Estimate state / orbital elements
3. Estimate associated uncertainty
4. Plan future observations
5. Update estimates with new data



## Sidereal time

- Measures Earth rotation relative to fixed stars (coordinate system)
- Equivalent to angle between x-axis (vernal equinox) and local meridian = right ascension
- Simplifies calculations of “where am I looking”



## Earth orbiting objects

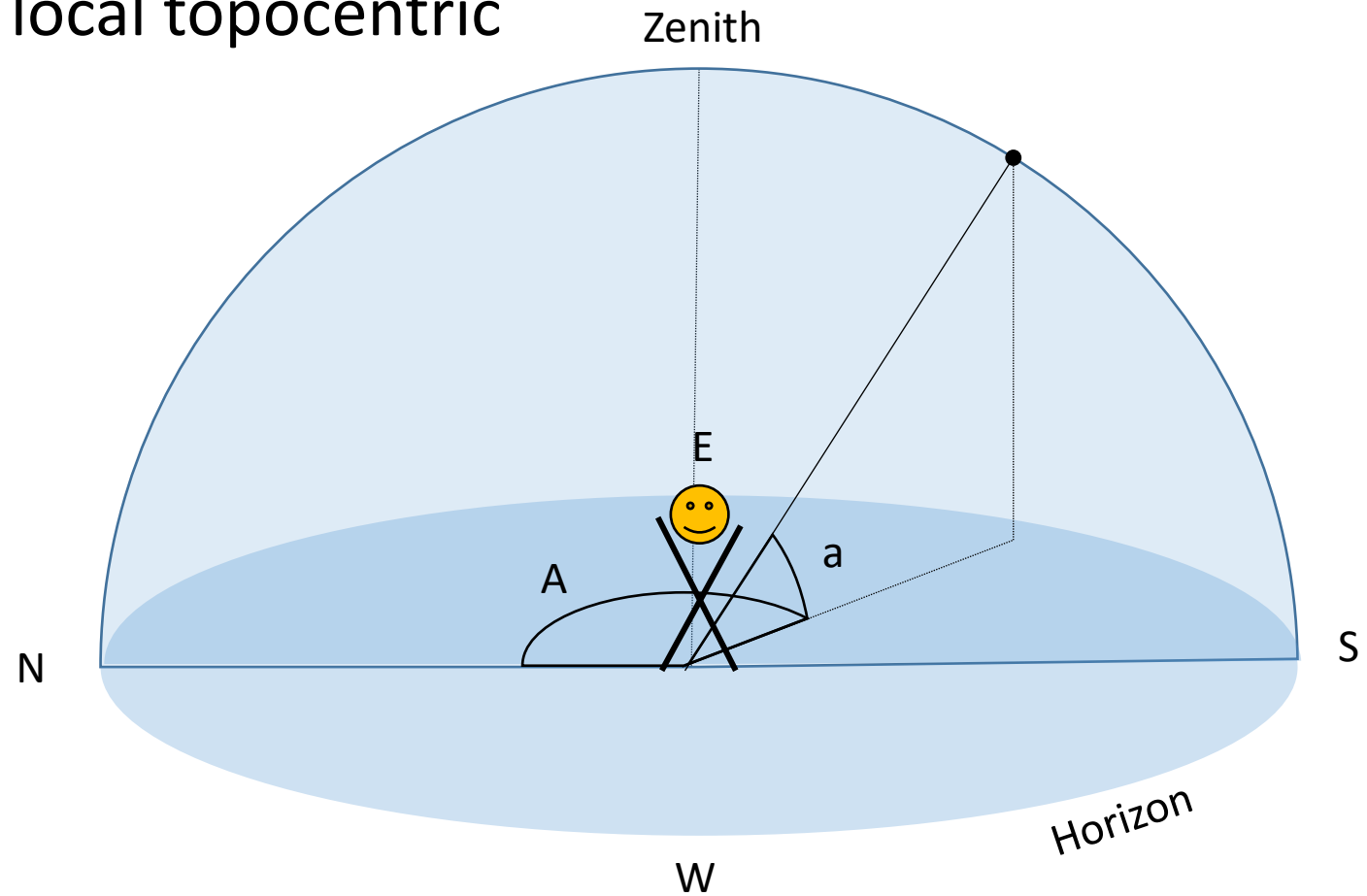
- Observation/Communication in local topocentric horizon frame

- Convert into ECI

- time
- obs. longitude
- obs. latitude
- ellipsoid (high accuracy only)

- ECI location representation

- Cartesian
- spherical



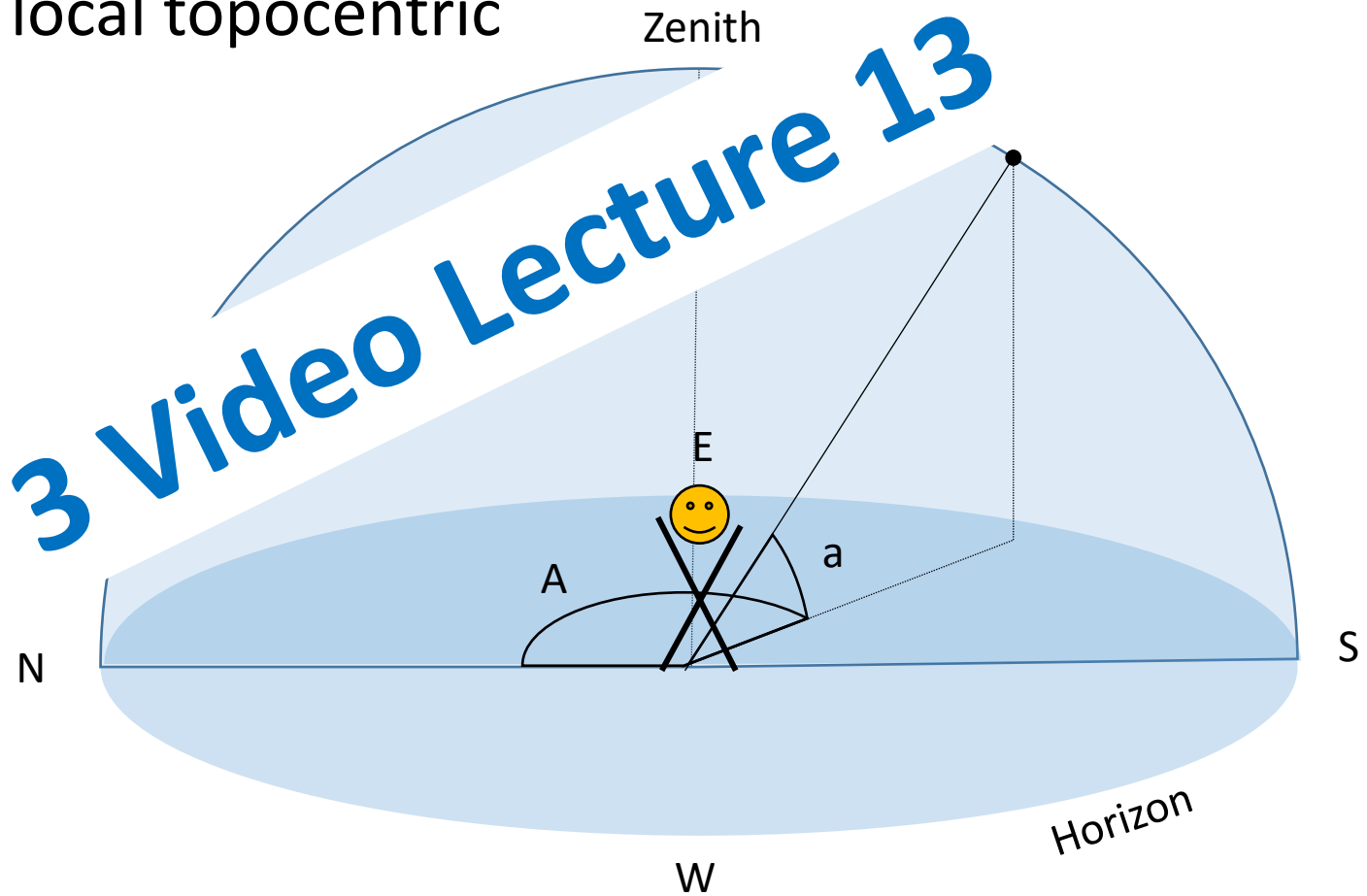


## Earth orbiting objects

- Observation/Communication in local topocentric horizon frame

- Convert into ECI
  - time
  - obs. longitude
  - obs. latitude
  - ellipsoid (high accuracy)

- ECI location
  - Cartesian
  - spherical



- Observations measured in local horizon frame
  - Angles, angular rates – optical observation
  - Range, range rate – radar observations
  - Time
- Conversion to Cartesian ECI / ecliptic
  - Position: from angles and range
  - Velocity: from additional angular & range rates
- Partial conversion
  - Direction in ECI without range





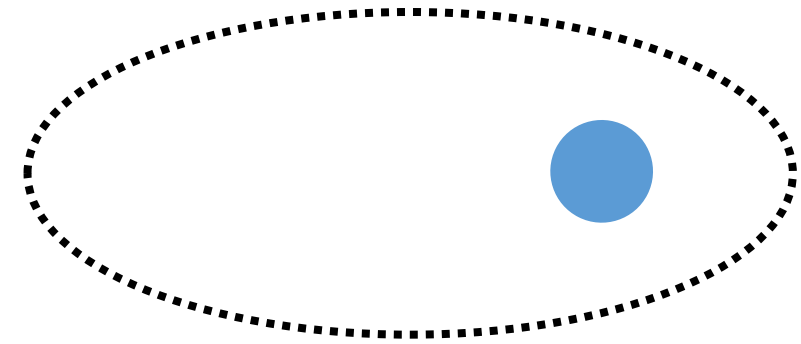
(Preliminary)

# Orbit Estimation & Determination

Taking a step back:

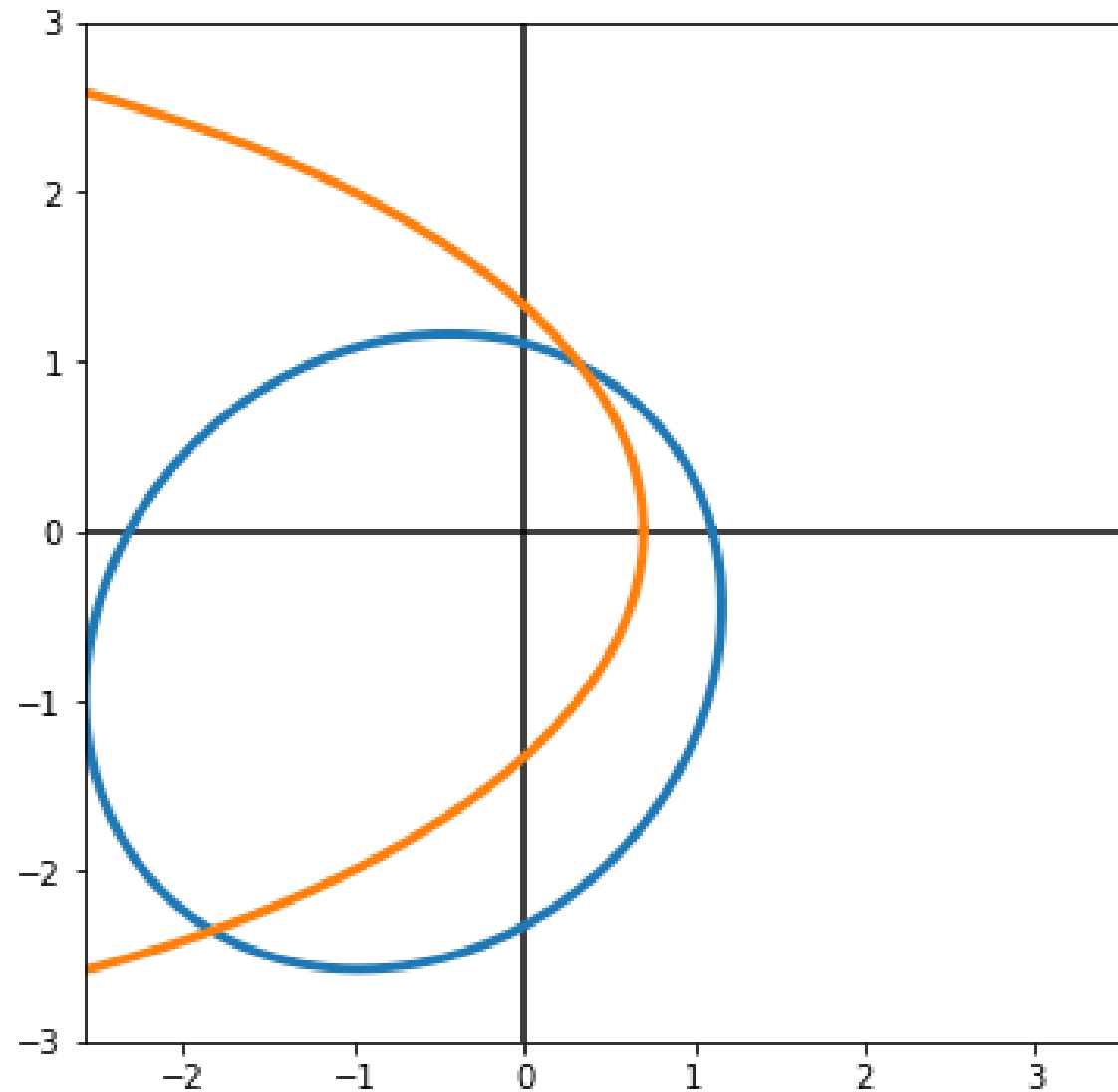
How many points are needed to determine a Keplerian orbit?

- Two points + primary define orbital plane
- Orbit shape given by  $a$ ,  $e$
- Two unknowns, two equations for orbit shape



$$r(\theta) = \frac{a(1 - e^2)}{1 + e \cos \theta}$$



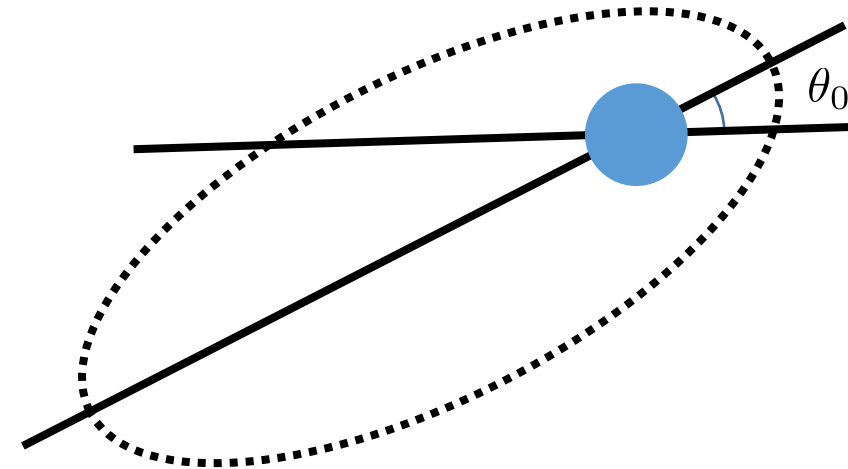


We'll see later:  
there are infinitely many  
orbits between two points!

Taking a step back:

How many points are needed to determine a Keplerian orbit?

- Two points + primary provide orbital plane
- Orbit shape given by  $a$ ,  $e$
- Three unknowns!
- Need **three points** → solve for orbit shape



$$r(\theta) = \frac{a(1 - e^2)}{1 + e \cos(\theta - \theta_0)}$$



- Depending on available data
- Depends on observation
- Many different methods available

We'll look at three classical methods:

- 3 positions:
- 2 positions + time:
- 3 sets of angles:

**Gibbs' method**

**Lambert's problem**

**Gauss' method**

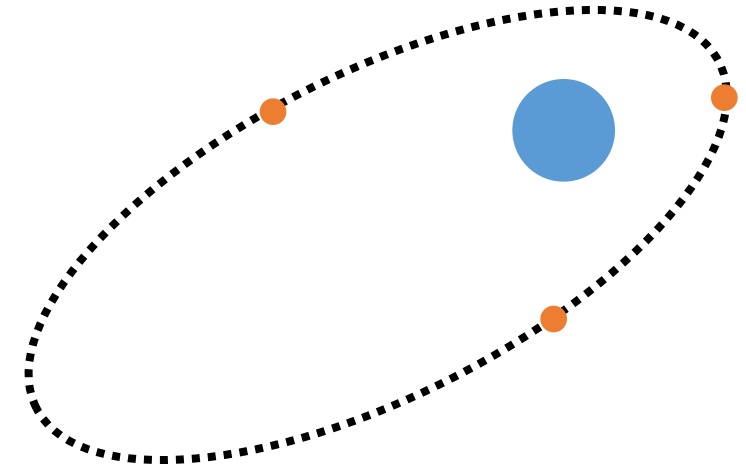


# Gibbs' method

3 distinct positions **in order**, to determine full orbit

$$\vec{r}_1, \vec{r}_2, \vec{r}_3$$

- Derived from e.g.
  - Combined Radar & Optical observations
  - GPS location
- Must be coplanar
  - not the case with observational data
- Geometrical
  - should be largely spaced





# Summary of Gibbs' method

1.  $\vec{N} = r_1(\vec{r}_2 \times \vec{r}_3) + r_2(\vec{r}_3 \times \vec{r}_1) + r_3(\vec{r}_1 \times \vec{r}_2)$

2.  $\vec{D} = \vec{r}_1 \times \vec{r}_2 + \vec{r}_2 \times \vec{r}_3 + \vec{r}_3 \times \vec{r}_1$

3.  $\vec{S} = (r_2 - r_3)\vec{r}_1 + (r_3 - r_1)\vec{r}_2 + (r_1 - r_2)\vec{r}_3$

4.  $\vec{v}_2 = \sqrt{\frac{\mu}{ND}} \left( \frac{\vec{D} \times \vec{r}_2}{r_2} + \vec{S} \right)$

5. Convert to Keplerian elements

Given:

$$\vec{r}_1, \vec{r}_2, \vec{r}_3$$

Wanted:

$$\vec{v}_2$$

Required:

$$\frac{\vec{r}_1}{r_1} \cdot \left( \frac{\vec{r}_2}{r_2} \times \frac{\vec{r}_3}{r_3} \right) = 0$$

(coplanar)

In order along track

For derivation see Chapter 5.2 in Curtis

# Summary of Gibbs' method

1.  $\vec{N} = r_1(\vec{r}_2 \times \vec{r}_3) + r_2(\vec{r}_3 \times \vec{r}_1) + r_3(\vec{r}_1 \times \vec{r}_2)$

2.  $\vec{D} = \vec{r}_1 \times \vec{r}_2 + \vec{r}_2 \times \vec{r}_3 + \vec{r}_3 \times \vec{r}_1$

3.  $\vec{S} = (r_2 - r_3)\vec{r}_1 + (r_3 - r_1)\vec{r}_2 + (r_1 - r_2)\vec{r}_3$

4.  $\vec{v}_2 = \sqrt{\frac{\mu}{ND}} \left( \frac{\vec{D} \times \vec{r}_2}{r_2} \right)$

5. Convert Cartesian elements

Given:

$\vec{r}_1, \vec{r}_2, \vec{r}_3$

Wanted:

$\vec{v}_2$

Required:

$$\frac{\vec{r}_1}{r_1} \cdot \left( \frac{\vec{r}_2}{r_2} \times \frac{\vec{r}_3}{r_3} \right) = 0$$

(coplanar)

In order along track

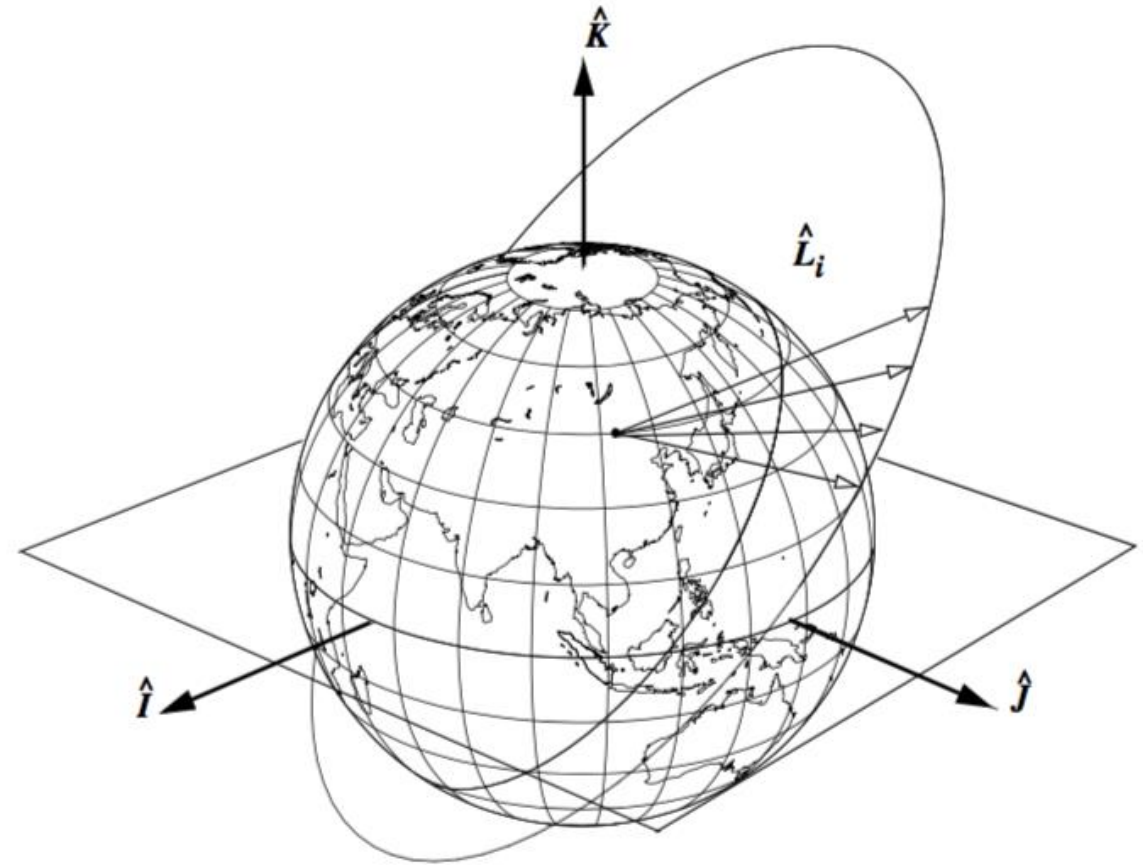
For derivation see Chapter 5.2 in Curtis

# Gauss' method



- Minimum of 3 sets of observation angles & times
- Orbit **estimation** method
  - Approximate
  - More inputs → better result (via e.g. least square fit)
  - Works best for nearby observations
- Original method devised by Gauss to find Ceres again
- Other (similar) methods exist

$$\alpha_1, \delta_1, t_1 \quad \alpha_2, \delta_2, t_2, \quad \alpha_3, \delta_3, t_3, \dots$$



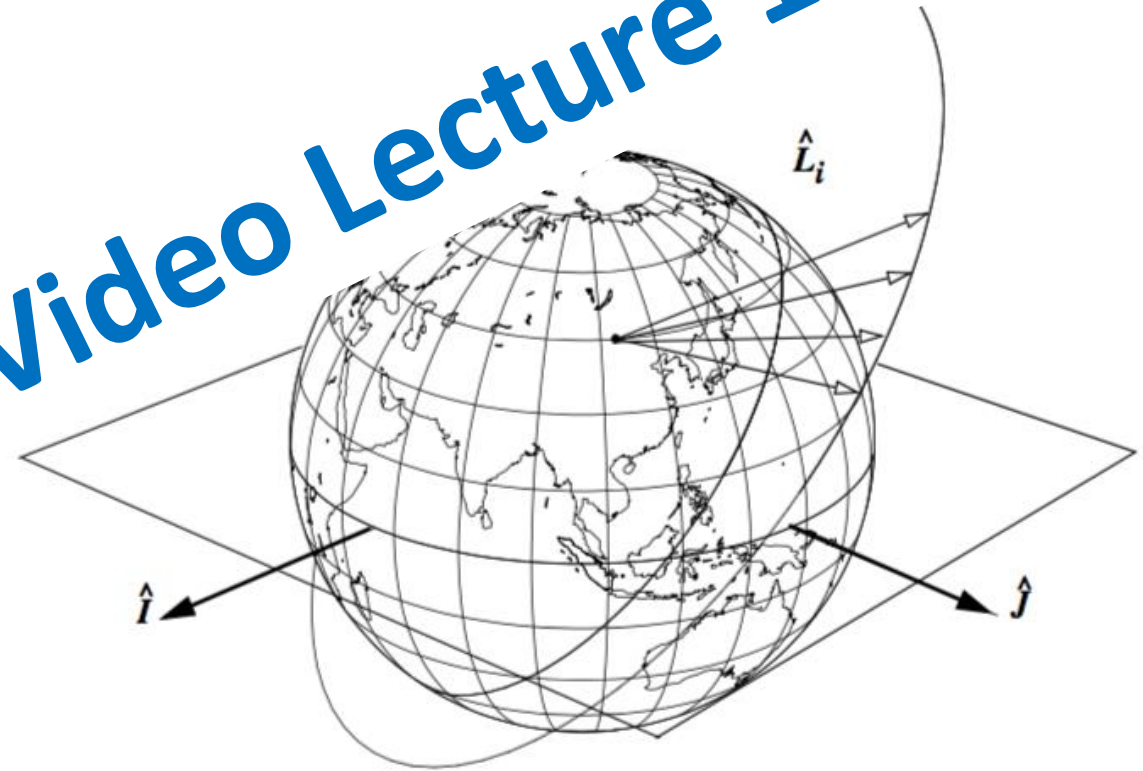
Vallado

**Figure 7-4. Geometry of Angles-only Observations.** The underlying principle for the angles-only technique is the use of topocentric angles to form line-of-sight vectors. For objects that are very distant, the distinction for topocentric values diminishes.

# Input to Gauss method

- Minimum of 3 sets of observation angles & times
- Orbit **estimation** method
  - Approximate
  - More inputs → better result (via e.g. least square fit)
  - Works best for near-Earth objects
- Original method used by Gauss to determine the orbit of Ceres again
- Other (similar) methods exist

$$\alpha_1, \delta_1, t_1 \quad \alpha_2, \delta_2, t_2, \quad \alpha_3, \delta_3, t_3, \dots$$



Vallado

**Figure 7-4. Geometry of Angles-only Observations.** The underlying principle for the angles-only technique is the use of topocentric angles to form line-of-sight vectors. For objects that are very distant, the distinction for topocentric values diminishes.

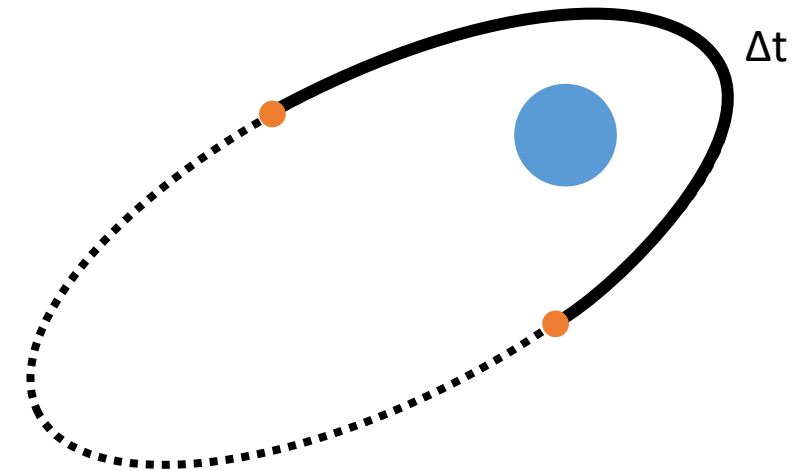
# Lambert's problem



## 2 positions and time between them

$$\vec{r}_1, \vec{r}_2, \Delta t$$

- 2 positions is not unique
- Additional transfer time yields unique orbit (with number of revs & direction)





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- Implement once, or use from library
  - Batin, general variables:  
modern, robust, suitable for numerics
- Or use numerical method
  - e.g. targeting with differential corrector in GMAT
- Applications in orbit determination and mission planning (targeting)

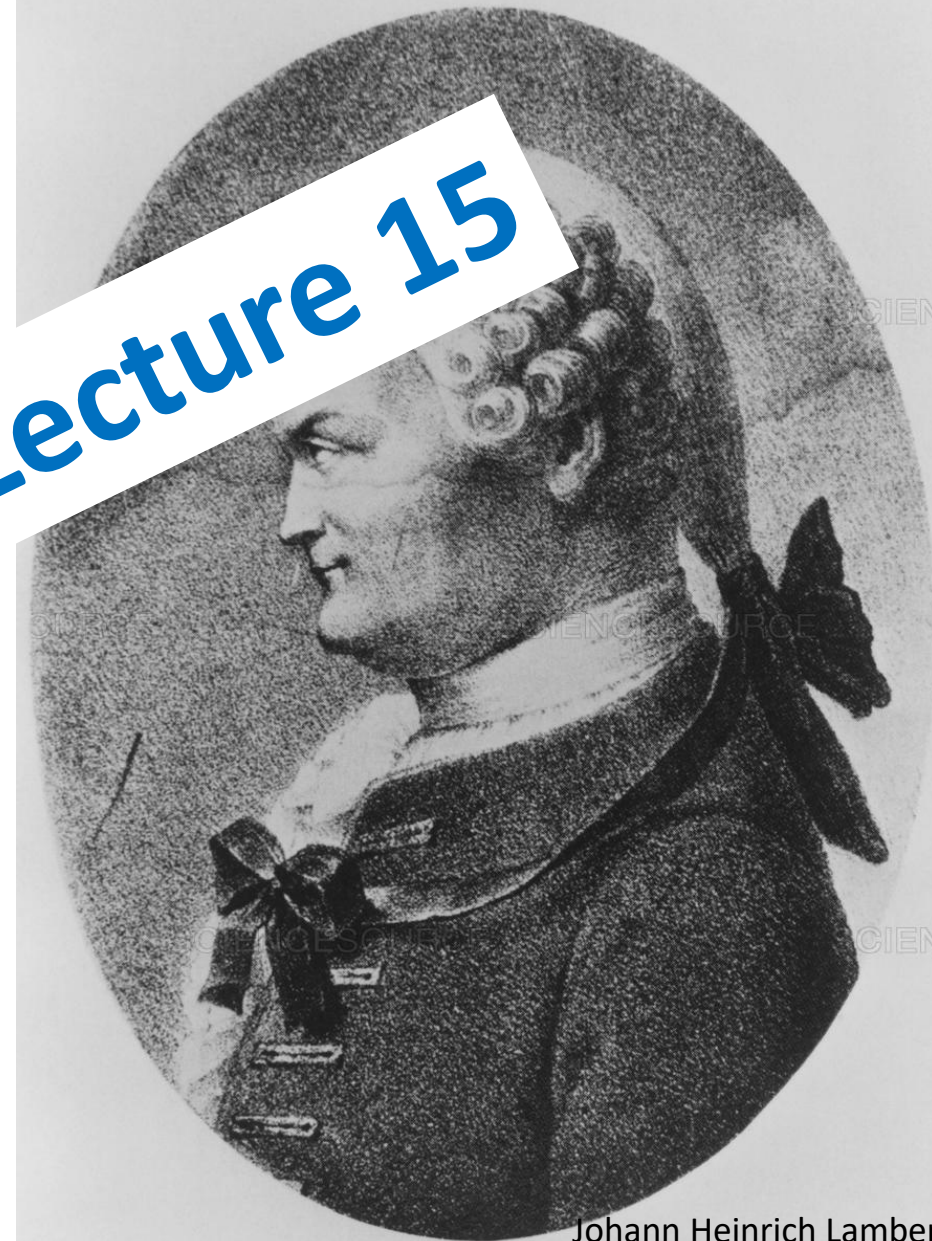


Johann Heinrich Lambert



- Implement once, or use from library
  - Batin, general variables:  
modern, robust, suitable for numerics
- Or use numerical method
  - e.g. targeting with differential correction
- Applications in orbit determination and mission planning

see Chapter 4 Video Lecture 15



Johann Heinrich Lambert

- Earth based observations are optical, laser, or radar, providing different measured quantities
- Single observation provides insufficient information
- Preliminary orbit determination combines observations to estimate an orbit
- Depending on available data, different techniques are used:
  - 3 positions (**Gibbs**)
  - 2 positions + time (**Lambert**)
  - 3 sets of angles (**Gauss**)

## Further reading

- Curtis, Chapter 5

Also:

- Vallado, Chapter 7
- Batin, Chapter 2

# Orbital Mechanics (SESA6076)

## Chapter 3: Orbit Determination

Dr. Alexander Wittig

